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ASSIGNMENTS - 3

Benefits of Serverless Architecture

- Serverless architecture allows developers to build and deploy applications without managing the underlying infrastructure. Using cloud services like AWS Lambda, Azure Functions, or Google Cloud Functions, code is executed in response to events and scaled automatically. This architecture reduces operational overhead, lowers cost by charging only for usage.

Progressive Web Apps (PWAs)

- Progressive Web Apps combine the best features of web and mobile applications. PWAs are accessible through browsers but provide a native-like experience with offline capabilities, push notifications, and home screen installation. They use service workers, manifest files, and responsive design to ensure fast, reliable performance across devices.

Role of AI & Machine Learning in Software

Architecture

- AI and ML enable system to analyse data, learn patterns & making intelligent decisions.

- Common use in e-commerce like,

- Personalized product recommendations

- Dynamic pricing adjustments

- Customer behavior prediction

- Architecturally AI/ML components are:

- Deployed as independent microservices

- Access via Cloud-based APIs

- This promotes modularity, separation &

Scalability & easy to maintain

Cloud Computing Service Models

- Software as a Service (SaaS): Deliver fully functional applications over the internet.

- Platform as a Service (PaaS): Provide a development with tools & frameworks.

- Infrastructure as a Service : offers virtualized computing resources . .